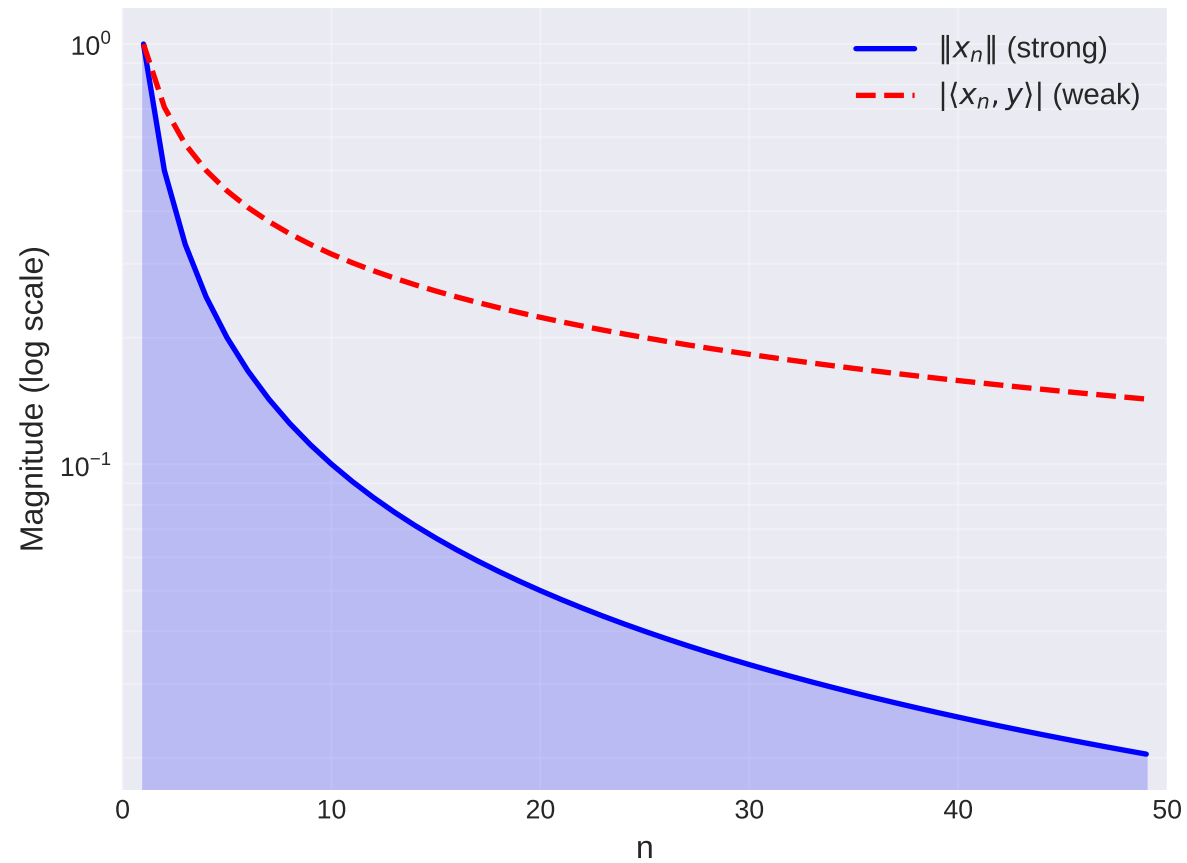


Strong vs Weak Convergence



Example: $x_n = \frac{1}{n} \sin(nx)$ in $L^2[0, \pi]$

- Converges weakly to 0
- Does NOT converge strongly to 0
- Oscillations prevent strong convergence